

- [illegible]

a) State how the elements **H** to **N** are arranged.

b) Why are elements **J** and **N** in the same group?

c) Why are elements **J** and **L** in the same period? (CLO1, C2)

- | Elements | Electronic configuration | Valence electronic configuration | Number of valence electrons | Period | Group | Block |
|--------------------|--------------------------|----------------------------------|-----------------------------|--------|-------|-------|
| ${}_8\text{O}$ | | | | | | |
| ${}_{20}\text{Ca}$ | | | | | | |
| ${}_{26}\text{Fe}$ | | | | | | |
| ${}_{34}\text{Se}$ | | | | | | |
| ${}_{29}\text{Cu}$ | | | | | | |

TABLE 1

Element	Period	Group	Valence Electronic Configuration	Electronic Configuration	Block
<i>A</i>	2	15			
<i>D</i>	4	6			
<i>E</i>	3	1			
<i>G</i>	4	17			

- (CLO3, C3)**

4. Arrange and explain the ascending order of the atomic radii for the elements below.
- Ne, Li, N
 - F, Cl, Br
 - Sc, Cr, Ni
- (CLO3, C3)

5. Table below shows the proton numbers of elements **P**, **Q**, **R**, **S** and **T** in the periodic table.

Elements	P	Q	R	S	T
Proton number	6	11	15	16	19

Arrange these elements in the ascending order of atomic size. Explain.

HOUR : 2

6. The table below shows the proton number and the stable ions of atoms **P**, **Q** and **R**.

Atom	P	Q	R
Proton Number	9	11	13
Ion	P ⁻	Q ⁺	R ³⁺

Arrange the ions in increasing order of ionic radius. Explain.

Compare the size of **P** and **Q** atoms with their respective ions. Explain.

PSPM 2002 / 2003 (CLO3, C4)

7.
 - Explain the term “isoelectronic species”
 - Arrange the following isoelectronic series in order of ascending radius. Explain.
F⁻, Ne, O²⁻, Mg²⁺, Na⁺
- (CLO3, C4)
8. **TABLE 2** below shows ionic radii of species in Period 3.

TABLE 2						
Species	Na ⁺	Mg ²⁺	Al ³⁺	P ³⁻	S ²⁻	Cl ⁻
Radius/nm	0.095	0.065	0.050	0.212	0.184	0.181

Explain the variation of ionic radii across Period 3.

(CLO3, C4)

9. Arrange the species below in a descending order of their radii.
- O²⁻, S²⁻, Cl⁻, Br⁻
 - Mg²⁺, Li⁺, K⁺, Ca²⁺
- (CLO3, C4)

HOUR : 3

10.
 - Define the first and second ionisation energy.
 - Write the equations to show the first ionisation energy and second ionisation energy of calcium. Compare and explain for the difference in these ionisation energies.
- (CLO3, C4)

11. **FIGURE 2** shows the variation in the first ionisation energy of the elements in the third period. Explain the trend across the period. **PSPM 2010 / 2011 (CLO3, C4)**

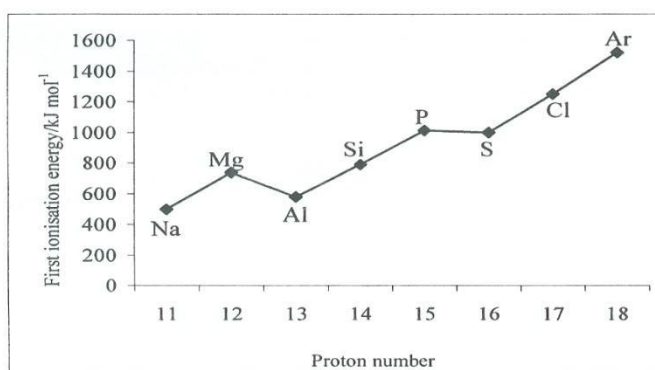


FIGURE 2

12. The table below shows the first ionisation energy of Group 1 elements.

Element	First ionisation energy (kJ mol ⁻¹)
Li	520.1
Na	495.8
K	418.8
Rb	402.9
Cs	375.6

Explain the trend of the first ionisation energy.

(CLO3, C4)

13. With reference to the table given below:

Element	¹¹ Na	¹² Mg
Ionisation energy	496 kJ mol ⁻¹	738 kJ mol ⁻¹

- Why is the first ionisation energy of sodium lower than that of magnesium.
- Which element has a higher second ionisation energy. Explain

PSPM Jan 2000 (CLO3, C3)

14. The electronic configurations of element **Q** and **R** are as follows:

Q	1s ² 2s ² 2p ⁶ 3s ² 3p ³
R	1s ² 2s ² 2p ⁶ 3s ² 3p ⁴

Between **Q** and **R**, which element has higher first ionisation energy? Give reason for each element.

15. Consider the following elements **P**, **Q**, **R** and **S** in **TABLE 3**.

TABLE 3

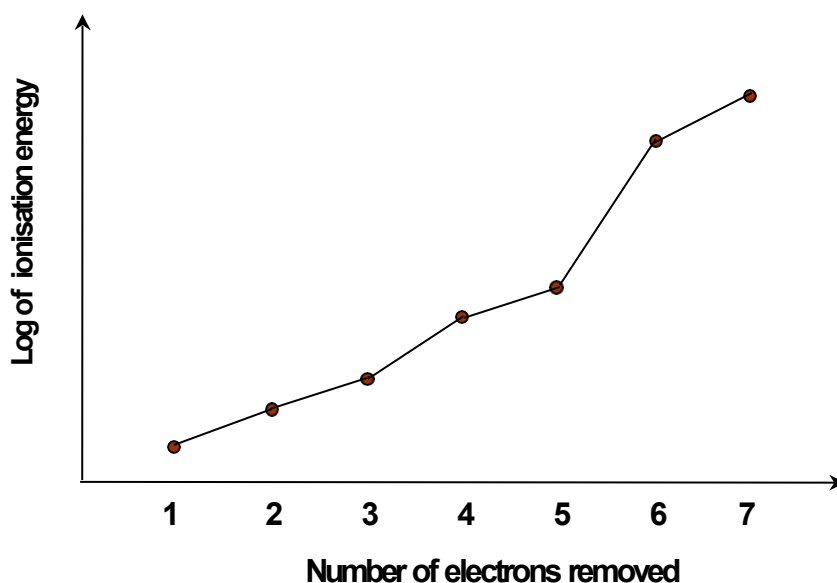
Element	Proton number
P	3
Q	7
R	11
S	12

- Determine group, period and block for element **R**.
- Choose elements with the same chemical properties.
- Which of these elements has the smallest atomic radius?
- Arrange elements in **TABLE 3** in the ascending order of first ionisation energy. Explain.

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HOUR : 4

16. The diagram below shows the energy required to remove the first seven electrons from an atom of element **Q**.



- Explain why the successive ionisation energies increase with the number of electrons removed.
- Explain the drastic increase in the ionisation energy of the sixth electron compared to that of the fifth electron.
- State the group where **Q** is located in the Periodic Table.
- Give the valence electronic configuration.
- If **Q** is in the third period, give the formula of the compound formed when **Q** atom combines with chlorine.

PSPM Jan 2000 (CLO3, C4)

17. **TABLE 4** shows the hypothetical values of successive ionisation energies of element **Y**.

TABLE 4

	IE 1	IE 2	IE 3	IE 4	IE 5	IE 6	IE 7
Ionisation Energy (kJ mol ⁻¹)	756	1400	2987	5000	24000	26000	28900

- Determine the group and block of element **Y**. Explain your answer.
- Write the electronic configuration of element **Y** if it is an element in period 3.

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